Group 11 Week 4 Report

1. Background/Theory

SAR

Synthetic Aperture Radar (SAR) is an active microwave sensor, meaning that it emits energy of its own and captures the energy reflected off the surface (backscatter). SAR does not require sunlight and microwaves penetrate clouds, allowing for the imaging of the ground in situations where optical sensors cannot. The backscatter also differs depending on the surface roughness, moisture and vegetation structure, and thus contains information which is not contained in optical reflectance.

Sentinel-1 supports dual Polarisation data in Interferometric Wide (IW) mode and works in C-band. VV (vertical transmit, vertical receive) is sensitive to surface scattering and soil scattering while VH (vertical transmit, horizontal receive) is sensitive to volume scattering from vegetation canopies. A Sentinel-1 Ground Range Detected (GRD) product is a calibrated and dB expressed product derived in Google Earth Engine (GEE). SAR is likely to enhance crop analysis when used in conjunction with optical information like NDVI in the presence of cloud cover.

Optical Features and NDVI

Optical satellite data gives information about the Earth's surface by sensing reflected solar radiation in specific wavelengths. For our project, we applied the existing optical data for Argentina and Brazil for researching the vegetation parameters and comparing corn, soybean, and wheat crops.

Sentinel-2 provides a variety of spectral bands, which reflect different features of vegetation and surface. The red band (B04) and the near-infrared band (B08) were used to calculate NDVI. Generally, vegetation absorbs red radiation and reflects near-infrared one, thus NDVI can be applied for studying vegetation greenness .

The formula for calculating NDVI is:

NDVI = B08-B04/B08+B04

where B08 stands for near-infrared band and B04 for red one. In our project, NDVI was calculated to explore the NDVI distribution for the existing optical samples and to compare three categories of crops.

2. Materials and Tools

Materials and Tools (SAR Extraction)

Datasets:

Google Earth Engine (COPERNICUS/S1_GRD) Sentinel-1 SAR GRD (IW mode, VV and VH polarisations, in dB). For each sampled pixel, used for the extraction of backscatter at 10 m.

Images from Sentinel-2 from COPERNICUS/S2_SR_HARMONIZED and COPERNICUS/S2_HARMONIZED from Google Earth Engine. For validating the field locations from the band values of the data set.

The data provided for Brazil and Argentina comes from YieldSAT field-level data (https://yieldsat.github.io/) in the form of two CSVs (Brazil_with_coordinates.csv, Argentina_with_coordinates.csv). They hold 100 pixel samples (50 per country) from 6 fields, with crop labels, optical band values, dates, pixel indices (row/col) and encoded coordinates (coord_x/y/z).

Tools

Google Earth Engine Code Editor (JavaScript): Making point geometries, filtering Sentinel-1 images, sampling VV and VH, visualising the two bands separately, and exporting the results.

The imported tables for Brazil and Argentina are stored as Earth Engine assets.

Python (pandas, NumPy): looking at the CSVs, decoding the coordinate vectors, and checking the exported table.

Google Drive: getting the exported CSV file.

Component	Details
Input data	Argentina.csv and Brazil.csv
Programming language	Python
Environment	Google Colab / Jupyter Notebook
Libraries	Pandas, NumPy, Matplotlib
Output file	Optical_Features.csv - optical feature dataset
	SAR_VV_VH_verified_pixels.csv - 50 valid SAR pixel samples

manifest.csv - field metadata and acquisition dates

SAR_field_level_features.csv - aggregated field-level SAR features

Week4_Optical_SAR_Merged.csv - preliminary merged dataset.

Week4_SAR_alignment_summary.csv - integration and data-quality summary

3. Methodology

SAR

1.) Location recovery: The provided CSVs were not provided with longitude/latitude, so during import GEE created empty geometries for all 100 records. The coord_x, coord_y and coord_z were determined to be unit-length vectors (all with norm 1.0 for all records), which is consistent with the following equations: $x = \cos(\varphi)\cos(\lambda)$, $y = \cos(\varphi)\sin(\lambda)$, $z = \sin(\varphi)$, where φ and λ are in degrees used as radians. This encoding is derived from the data, but not described in the dataset source.

2.)Validation: Decoding leaves repeated candidate locations every 6.28° of longitude, therefore, each field's location was chosen by comparing the Sentinel-2 values at the decoded locations with the values in the dataset. In Brazil field 10, there were 4 bands (B2, B3, B4, B8) that were exact matches. There was a strong correlation between the 16-pixel NIR pattern and Argentina field 468 (0.93), and Argentina field 52 provided close band values.

3.)Geometry: Point geometries were created for the 50 pixel samples of the three verified fields (17 Brazil, 33 Argentina), and all are non-empty. The 100 records are actually pixel samples (row/col) of 6 fields, not 100 separate fields.

4.)Sentinel-1 filtering: COPERNICUS/S1_GRD images were filtered by location, IW mode and VV and VH. The one image in each field most like the date of the dataset (within ±12 days) was chosen; this ensures a consistent direction of orbit within each field.

5.)Extraction: Sampling was performed at 10 m scale, at every pixel, of VV and VH in dB. If the value of the variable was missing, it was retained as missing and identified (valid_VV, valid_VH, status). Zero was never used to fill in missing values. These scenes are in UTM projection (latest code: EPSG:32722 for Brazil; EPSG:32720 for Argentina), and have a nominal pixel size of 10 m.

Optical Feature Extraction

The first step was loading the Argentina and Brazil CSV datasets in Python. The data records, crops, missing values, and duplicates were observed for an analysis of data quality.

Next, after observing the data quality, we computed NDVI using red and near-infrared bands. For each record, the NDVI calculation formula was used to calculate NDVI value, which was added as a new column in the dataset.
The workflow follows:

- Argentina and Brazil datasets
- NDVI Calculation
- Processed features
- Generated visualisations

The processed data was saved in the csv. Next, we created three plots - NDVI histogram for looking at the distribution of NDVI value, box plot of crops for analyzing three types of crops, and box plot of optical bands for looking at selected spectral bands.
The dataset alignment and validation phase was carried out to evaluate the representativeness and consistency of the selected optical datasets: images from the YieldSAT, in terms of the precision and saturation with which they could be used in the workflow of the crop-classification. The csv files were analyzed by using Python and Panda.
The entire validation process involves:

- Review of Recorded Numbers
- Unique Field Identifiers
- Crop Class Distribution
- Coordinates check
- Observation dates

SAR Preprocessing and Optical - SAR Integration

The 50 valid Sentinel-1 pixel samples were grouped by country, farm identifier, and field identifier to calculate field-level mean and standard deviation for VV and

VH backscatter. The no. of pixel samples contributing to each field summary was also recorded

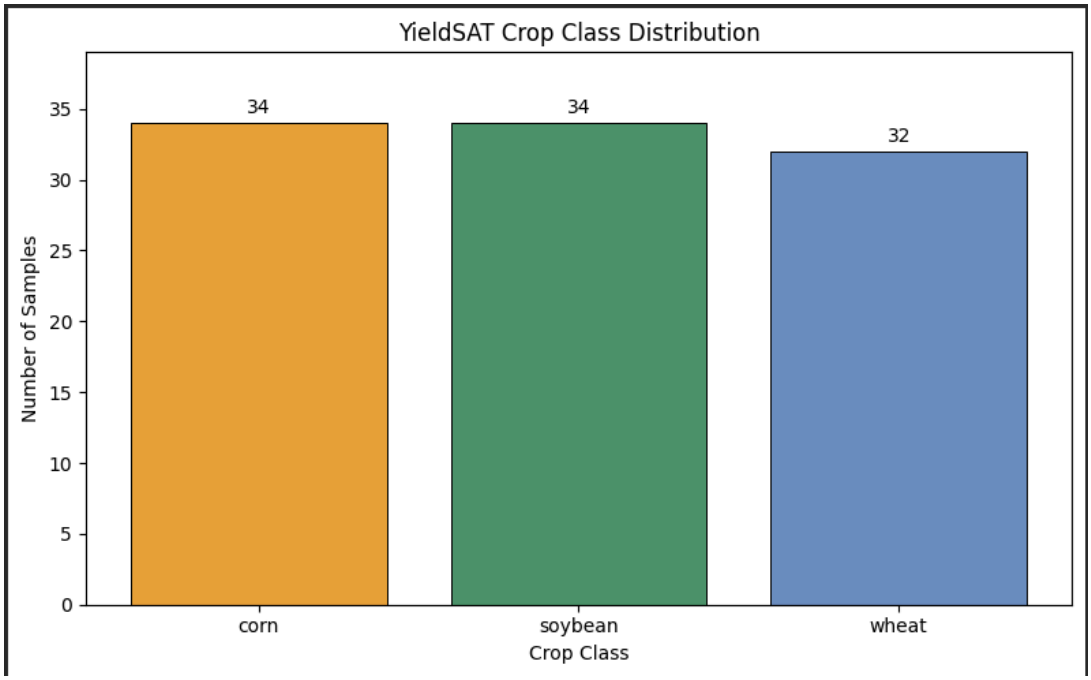
The resulting SAR feature table was merged with the optical dataset using the corresponding country, farm identifier, and field identifier. Left join was used to keep all optical observations and identify records without corresponding SAR features. Missing SAR values were retained rather than replacing them with artificial values. The merged dataset was then checked for matching fields, unmatched observations, and missing SAR features.

4. Results - Optical Data Analysis

The results were summarized into a preliminary validation report and downloaded as a csv file. In the preliminary validation, 50 records were investigated from Argentina and 50 records were investigated from Brazil, resulting in a total of 100 records. There were 3 different field identifier samples in each country. No missing field identifiers, invalid observation dates or missing coordinate values were found for the samples examined.

In total 34 corn records, 34 records for soybeans, and 32 wheat records were seen among the crop class distribution. This distribution suggests the selected sample size was normal and included all 3 crop classes with relatively similar sample sizes.

The distribution of the three classes of crops in the 100-record sample is illustrated in the below figure.



The results collaborate the completeness of the metadata fields inspected. However, the checks are indicative of the data validation prior to validating optical/SAR successful alignment. For correlation Sentinel-1 should be further validated with reference to optical observations acquired at comparable dates, places and spatial resolutions.

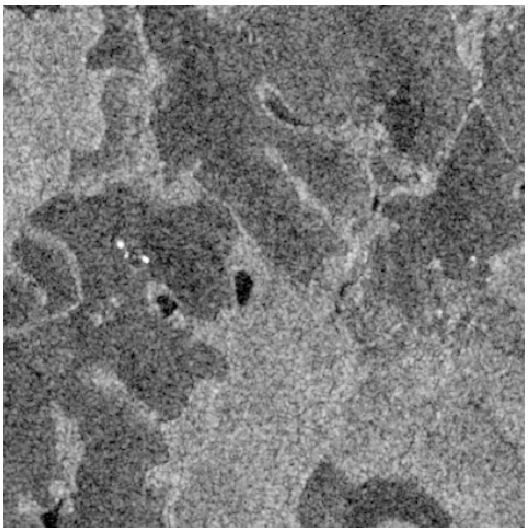
Our first analysis showed changes in NDVI values and clear differences between the crop types. This gave us a sense of what the optical data can tell us. Still we have to check on the accuracy of crop classification.

Updated SAR Results for the Team

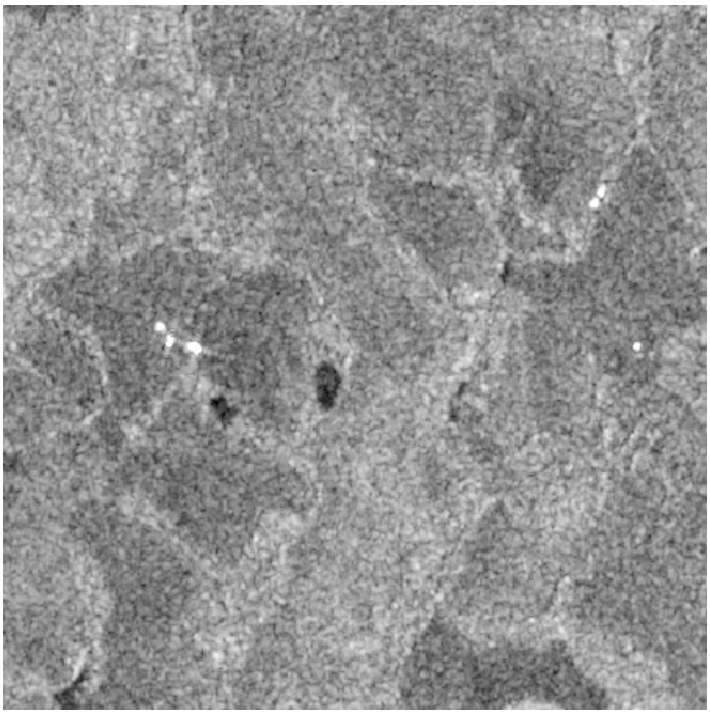
The dataset has a total of 100 pixel samples, with 50 being found and retrieved. These are from three fields: Brazil field 10, Argentina fields 52 and 468. No missing, masked or zero values were returned for all 50 samples for VV and VH values from Sentinel-1.We got VV and VH values for 50 pixel samples from Argentina and Brazil. The average VV value came out to about −10.123 dB. The average VH value was −16.813 dB. We still need to confirm that these SAR samples line up with the optical dataset before we proceed with feature fusion. VV averaged −8.88 dB (range −10.46 to −5.81) and VH averaged −15.23 dB (range −18.11 to −11.98). In Argentina field 52 the band values were near the

dataset values, as in the near-infrared band, which had values of 3068 and 3062, respectively.

Brazil field 10 (2022-10-20, descending):

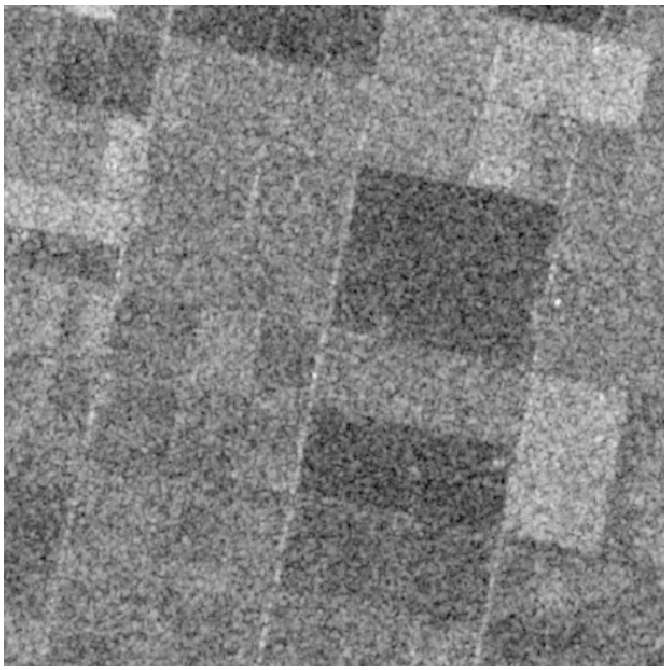


VV (dB), display range -25 to 0

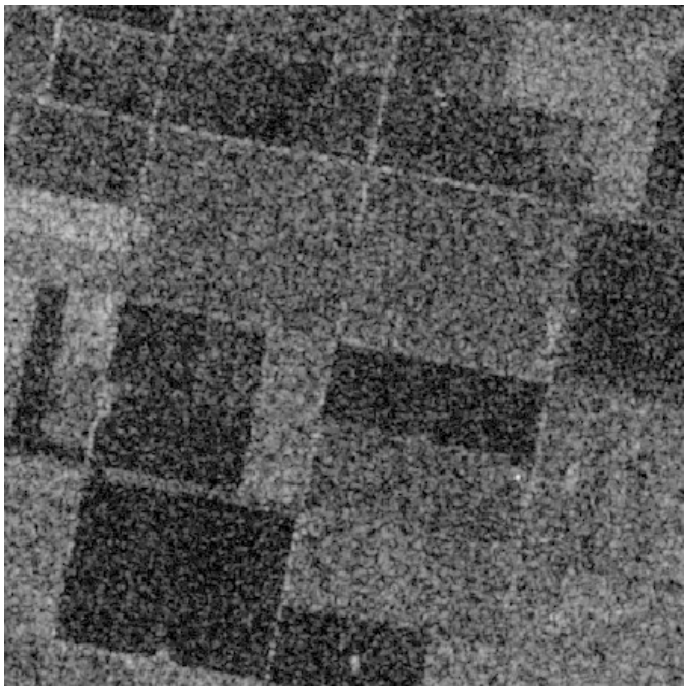


VH (dB), display range -30 to -5

Argentina field 52 (2021-10-25, descending):

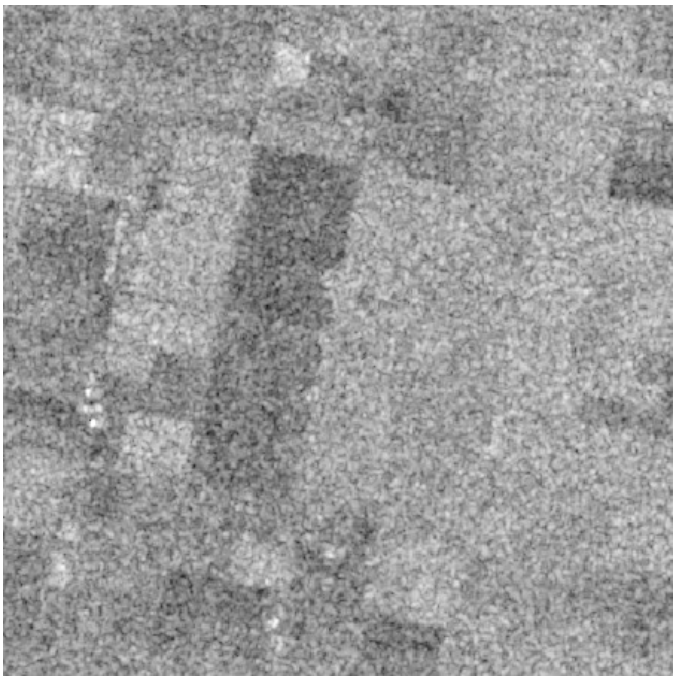


VV (dB), display range -25 to 0

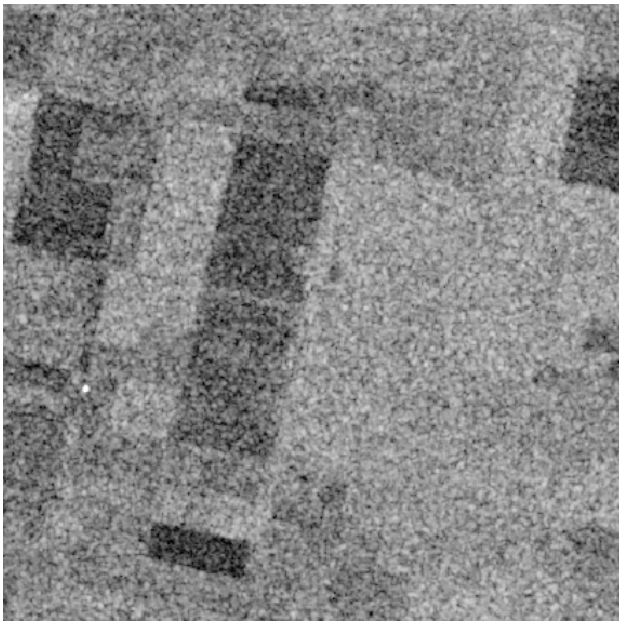


VH (dB), display range -30 to -5

Argentina field 468 (2021-04-16, descending):



VV (dB), display range -25 to 0



VH (dB), display range -30 to -5

Preliminary Optical - SAR Integration Results

The optical dataset contains 100 observations with 6 unique fields, while the Sentinel-1 dataset contains 50 valid pixel samples with 3 unique fields. The mean VV and VH backscatter values were -10.123 dB and -16.813 dB.

After aggregation, the SAR data contained 3 field-level records. The optical-SAR merge retained all 100 optical observations. SAR features were available for 50 optical observations matching the 3 fields, while the other 50 observations lacked matching SAR features.

The matched fields were Argentina farm 7, field 52; Argentina farm 38, field 468; and Brazil farm 1, field 10. The unmatched fields were Argentina farm 39, field 557; Brazil farm 3, field 96; and Brazil farm 3, field 117.

These results demonstrate preliminary integration for 3 fields, but complete optical-SAR feature coverage has not yet been achieved.

5. Discussion:

The Week 4 results show progress in preparing optical and SAR features for agricultural analysis. All 50 extracted SAR pixel samples contained valid VV and VH measurements, which were aggregated into field-level features for 3 unique fields.

The preliminary integration retained all 100 optical observations, but only 3 of the 6 unique fields had matching SAR features. Consequently, half of the optical observations didn't have SAR measurements in the merged table. This prevented the current dataset from being treated as a fully aligned optical-SAR dataset.

The results also highlight the importance of verifying field identifiers and acquisition dates before combining satellite observations. The manifest contains coordinate and CRS information that requires further validation, and an acquisition-date/year discrepancy is yet to be checked against the source metadata.

The next step should investigate the 3 unmatched fields, validate temporal consistency, and prepare separate optical-only, SAR-only, and fused datasets. Then, after field-level alignment and feature coverage have been verified, we will do model development.

6. Learnings:

- We discovered how to programatically review satellite data and ensure completeness in the field identifiers, observation date, and coordinate data.
- We learned how to calculate NDVI using near-infrared bands.
- We also discovered how to use histograms and box plots to see data patterns and compare crop categories. Also SAR data must line up before their features can be mixed.
- Experience was gained in using Python to generate an initial validation report and creating visualisations of the distribution of crop classes.
- Checking the completeness of the metadata was a different question from the actual spatial and temporal periodicity of SAR and optical satellite observations.
- Don't worry about missing geometry, that's not missing data. We were expecting a record to be missing, but none was found in Earth Engine, but there was still a record present in the encoded unit vectors. We learned how to check what columns represent on a table before we deem it unusable.
- There must be independent data to test an inferred transformation. The coordinates were then translated and a number of possible locations were suggested, the only way to discern which was correct was to compare them with the actual Sentinel-2 values. If it wasn't for that check we would have taken SAR from the incorrect location and not realized it.
- It is unlikely that the same sensor will observe on the same day. The Sentinel-1 nearest date used was 2.6 to 5.6 days before or after the dataset date, so both dates are presented in the output and we mentioned the matching rule, so that the SAR and optical features are in the same game later in the output.
- We learned to aggregate pixel-level VV/VH measurements into field-level statistical features and assess missing values during data integration.
- We learned that a successful table merge does not guarantee complete multimodal coverage. The field identifiers and acquisition dates must be validated before feature fusion.

7. **References**:

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